

Intune Mobile Security Protection Framework

Summary

Microsoft's mobile protection and configuration framework has been available for a considerable period and provides well-established guidance for applying layered security controls to Android Enterprise and iOS/iPadOS. However, translating that guidance into a complete and consistent Intune configuration has traditionally required substantial technical effort. Organizations have had to interpret the individual recommendations, create numerous app protection, compliance, and device configuration policies, align settings across platforms and protection levels, and prepare assignments and rollout rings before devices could be enrolled against the intended baseline.

This publication makes the framework significantly easier to adopt. It packages the target policy set with a PowerShell deployment script that rapidly creates the required Intune policies and makes the baseline available for validation, assignment, and device enrollment. The script is an implementation tool rather than the framework itself: the value of the publication lies in making an established Microsoft-aligned security outcome practical to deploy without requiring administrators to build every policy manually.

[Levels of protection and configuration in Microsoft Intune](#)

The framework aligns mobile controls to three complementary outcomes. App Protection Policies contain organizational data within managed applications. Device Compliance Policies establish whether a managed device is trusted enough to access corporate resources. Device Configuration Policies apply platform restrictions and security settings that reduce the attack surface. Together, these controls create a consistent security posture that can be enforced through Microsoft Entra Conditional Access.

[Data Protection Framework Using App Protection Policies](#)

The figure below summarizes the framework's policy composition across policy category, protection level, and platform family.

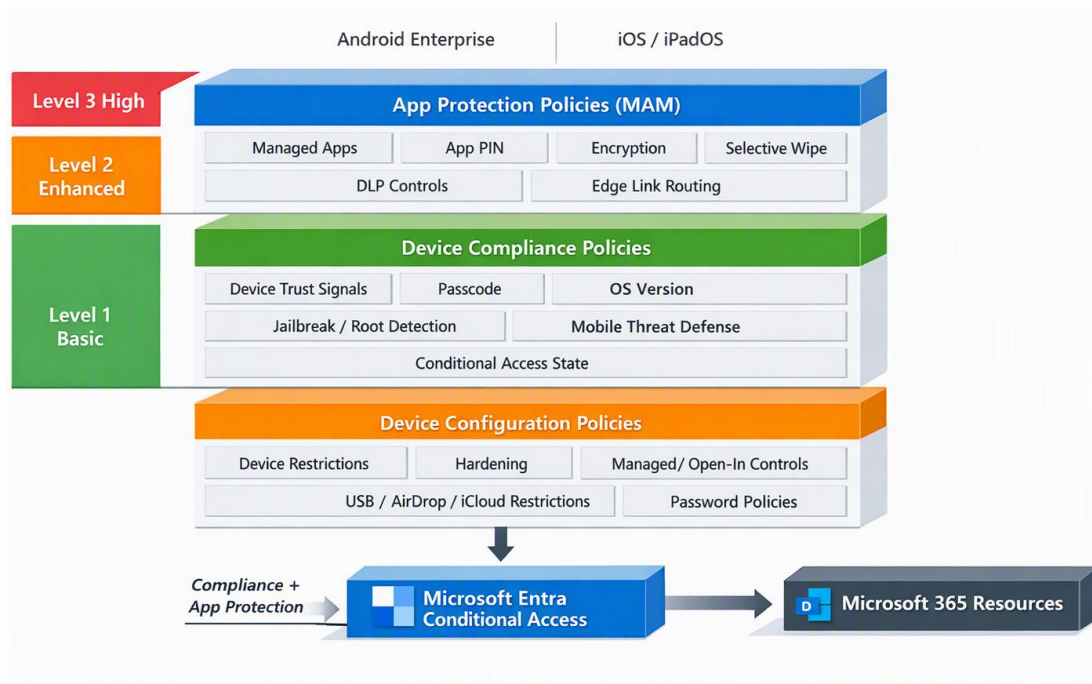


Figure 1: Layered view of the imported Intune policy set: app protection contains corporate app data, compliance evaluates device trust for access decisions, and configuration policies harden enrolled Android Enterprise and iOS/iPadOS devices across the three Microsoft security levels.

Resulting Security Architecture

The framework creates a three-layer mobile security architecture. App Protection Policies apply data-loss-prevention and access controls to managed applications, including scenarios where a device is not enrolled in Intune. Device Compliance Policies assess whether enrolled devices satisfy defined trust requirements and provide that status to Microsoft Entra Conditional Access. Device Configuration Policies harden enrolled devices by restricting risky operating-system capabilities, controlling data transfer paths, enforcing password behavior, and limiting unmanaged storage or sharing.

[Learn about Conditional Access and Intune](#)

[Use compliance policies to set rules for devices you manage with Intune](#)

[How to Create and Assign App Protection Policies](#)

The following diagram illustrates the intended policy stack and how the three security levels apply across MAM, compliance, and configuration.

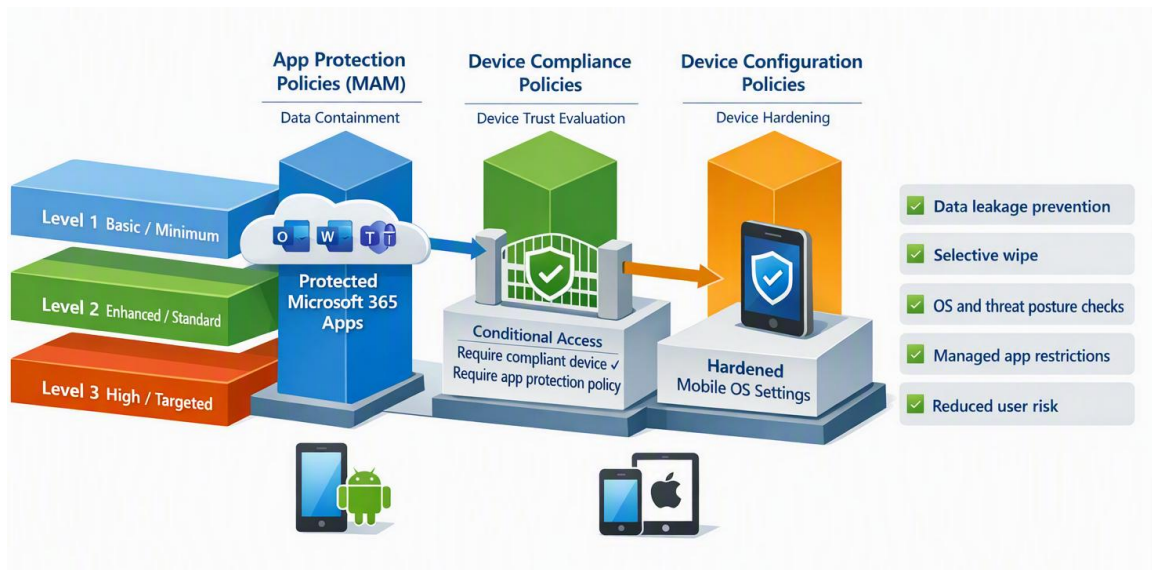


Figure 2: The framework creates a layered mobile security model in which app protection contains corporate data, compliance evaluates device trust for Conditional Access, and configuration policies harden Android Enterprise and iOS/iPadOS devices across Level 1, Level 2, and Level 3.

Protection Levels Within the Framework

Microsoft Intune documents three protection/configuration levels: Level 1 - Minimum protection and configuration, Level 2 - Enhanced protection and configuration, and Level 3 - High protection and configuration. Microsoft explicitly states these levels are starting points that organizations should customize, and even mix across policy types — for example, using Level 1 device configuration with Level 3 app policies if that matches business need.

Level	Microsoft intent	Typical target population	Security tradeoff
Level 1 / Basic / Minimum	Minimum baseline every organization should have; for app protection, it requires PIN, encryption, selective wipe, and Android attestation.	Broad mobile population, initial rollout, lower-risk access scenarios	Lowest user friction; limited DLP restrictions
Level 2 / Enhanced / Standard	Standard protection for devices where users access sensitive or confidential information; applicable to most mobile users accessing work or school data.	Majority of enterprise users, knowledge workers, managed BYOD and corporate mobile users	More DLP and OS requirements; moderate usability impact
Level 3 / High / Targeted	High protection for sophisticated security programs, uniquely targeted users, or users handling highly sensitive data where disclosure could materially harm the organization.	Executives, privileged admins, legal/finance/M&A, regulated data handlers, high-risk geographies or roles	Strongest restrictions; highest helpdesk and usability impact

Microsoft recommends a ring-based deployment approach for the App Protection framework and also recommends evaluating threat environment, risk appetite, and usability impact before applying Android and iOS/iPadOS security configuration examples broadly.

Policy Catalog: Framework Components

The following catalog groups the framework components by category, platform, and protection level. Each entry describes the operational and security outcome delivered by the policy rather than its underlying implementation syntax.

#	Policy	Category	Platform / ownership	Level	Operational and security outcome
1	Android Enterprise Basic Data Protection v1.7	App Protection	Android	L1	Establishes basic app PIN, encryption, Android attestation, device compliance requirement, and selective wipe posture for Core Microsoft apps.
2	iOS/iPadOS Enterprise Basic Data Protection v1.2	App Protection	iOS/iPadOS	L1	Establishes basic app PIN, app data encryption when device is locked, compliance requirement, and selective wipe posture.
3	Android Enterprise Enhanced Data Protection v1.9	App Protection	Android	L2	Adds DLP: managed-app outbound transfer, save-as block, Edge for links, notification redaction, screen capture block, Android minimum OS, and hardware-backed attestation.
4	iOS/iPadOS Enterprise Enhanced Data Protection v1.5	App Protection	iOS/iPadOS	L2	Adds managed-app outbound transfer, save-as block, Edge for links, notification redaction, iOS minimum OS, and disabled-account block behavior.
5	Android Enterprise High Data Protection v1.9	App Protection	Android	L3	Adds inbound managed-app restriction, simple PIN block, 6-digit PIN, annual PIN reset, print block, MTD secured threshold, data ingestion block, and approved-keyboard control.
6	iOS/iPadOS Enterprise High Data Protection v1.5	App Protection	iOS/iPadOS	L3	Adds inbound managed-app restriction, simple PIN block, 6-digit PIN, annual PIN reset, print block, third-party keyboard block, MTD secured threshold, and custom dialer requirement placeholder.
7	Fully managed basic security compliance (Level 1) v1.2	Compliance	Android fully managed	L1	Requires baseline device trust: Play integrity / certification, Android 9.0 minimum, password, encryption, and Intune app integrity.

#	Policy	Category	Platform / ownership	Level	Operational and security outcome
8	Fully managed enhanced security compliance (Level 2) v1.2	Compliance	Android fully managed	L2	Adds password expiration and previous-password history to the fully managed baseline.
9	Fully managed high security compliance (Level 3) v1.1	Compliance	Android fully managed	L3	Raises minimum OS to Android 11.0 and requires secured device threat level / advanced threat protection.
10	Work profile enhanced security compliance (Level 2) v1.4	Compliance	Android personal work profile	L2	Blocks rooted devices, unknown sources, USB debugging, requires encryption, Play services, security providers, Company Portal integrity, and hardware-backed evaluation.
11	Work profile high security compliance (Level 3) v1.2	Compliance	Android personal work profile	L3	Adds Android 11.0 minimum, password expiration/history, and secured threat protection requirement.
12	iOS/iPadOS enhanced security compliance (Level 2) v1.2	Compliance	iOS/iPadOS	L2	Blocks jailbroken devices, requires passcode, blocks simple passcodes, enforces iOS 14.8 minimum and immediate noncompliance action.
13	iOS/iPadOS high security compliance (Level 3) v1.3	Compliance	iOS/iPadOS	L3	Raises minimum OS to iOS/iPadOS 15.0 and requires secured threat protection / Defender risk posture.
14	Fully managed basic security configuration (Level 1) v1.2	Configuration	Android fully managed	L1	Hardens corporate Android devices with password, encryption-supporting posture, app updates, threat scan, USB/external media block, and factory reset block.
15	Fully managed enhanced security configuration (Level 2) v1.2	Configuration	Android fully managed	L2	Adds Google-account block, user add/remove block, password expiration/history, and stricter wipe threshold.
16	Fully managed high security configuration (Level 3) v1.2	Configuration	Android fully managed	L3	Adds always-on app updates, tethering/NFC/date-time restrictions, lock-screen feature restrictions, and stronger password/wipe controls.

#	Policy	Category	Platform / ownership	Level	Operational and security outcome
17	Work profile enhanced security configuration (Level 2) v1.3	Configuration	Android personal work profile	L2	Separates work/personal data with copy/paste block, screen capture block, account-add block, work profile password, and managed work profile restrictions.
18	Work profile high security configuration (Level 3) v1.2	Configuration	Android personal work profile	L3	Adds work notification block, contact-search block, widget block, trust-agent block, password expiration/history, and lower wipe threshold.
19	iOS/iPadOS Personal basic security configuration (Level 1) v1.1	Configuration	iOS/iPadOS personal	L1	Enforces baseline passcode, encrypted backup, iCloud managed-app sync block, lock-screen privacy, AirDrop unmanaged target, and untrusted certificate block.
20	iOS/iPadOS Supervised basic security configuration (Level 1) v1.1	Configuration	iOS/iPadOS supervised	L1	Adds supervised-only hardening such as AirPrint protections, proximity/password sharing blocks, keychain sync block, and activation lock allowance.
21	iOS/iPadOS Personal enhanced security configuration (Level 2) v1.2	Configuration	iOS/iPadOS personal	L2	Adds managed document transfer controls, diagnostic data block, enterprise book backup/sync block, and on-device dictation/translation enforcement.
22	iOS/iPadOS Supervised enhanced security configuration (Level 2) v1.2	Configuration	iOS/iPadOS supervised	L2	Adds supervised DLP controls, iCloud document sync block, USB Files access block, managed pasteboard requirement, and enterprise book controls.
23	iOS/iPadOS Personal high security configuration (Level 3) v1.2	Configuration	iOS/iPadOS personal	L3	Adds Handoff block, screen capture block, AirPlay pairing requirement, Apple Watch auto-unlock block, password expiration/history, and stricter wipe threshold.
24	iOS/iPadOS Supervised high security configuration (Level 3) v1.2	Configuration	iOS/iPadOS supervised	L3	Adds strongest supervised restrictions: App Store block, AirDrop block, iCloud backup/document sync block, Siri/iTunes/game restrictions, VPN creation block, app removal block, profile-

#	Policy	Cate- gory	Platform / owner- ship	Level	Operational and security outcome
					change block, and account modification block.

App Protection Policies in Depth

The six App Protection Policies implement Microsoft's Data Protection Framework using App Protection Policies. Their outcome is consistent protection of organizational data inside managed apps, regardless of whether the device is enrolled. Controls are organized around data protection and relocation, access requirements, and conditional launch, covering transfer restrictions, encryption, PIN and biometric access, operating-system checks, compromised-device detection, offline grace periods, and selective wipe.

Level 1 / Basic Data Protection applies to Android and iOS/iPadOS and is designed as an introductory baseline: work data is protected by app PIN, encryption, compliance checks, Android device attestation, and selective wipe capability while allowing broad data transfer to all apps and allowing printing/backup behavior. This level is appropriate for broad enablement when the organization wants app-layer protection with minimal user disruption.

Level 2 / Enhanced Data Protection adds the practical DLP controls Microsoft recommends for most mobile users accessing work or school data: outbound transfer is limited to managed apps, save-as is blocked except approved services, web links are directed to Microsoft Edge, notifications hide organization data, and minimum OS versions are required. On Android, the imported policy additionally blocks screen capture and uses hardware-backed evaluation; Microsoft notes hardware-backed evaluation provides stronger integrity signals but should be tested before broad rollout.

Level 3 / High Data Protection is for high-risk users or highly sensitive data and adds advanced data containment: receiving data only from managed apps, printing blocked, stronger app PIN requirements, MTD secured threshold, data ingestion restrictions, keyboard restrictions, and tighter handling of dialer/telecommunication flows. Microsoft states Level 3 can significantly affect users or apps and is intended for organizations or users facing targeted, sophisticated adversaries.

Operationally, these APP policies should be paired with Conditional Access. Microsoft Learn states that Conditional Access is recommended with APP to ensure policies are enforced, and the Data Protection Framework specifically calls out Conditional Access policies to require approved client apps or app protection policy for work or school data.

Device Compliance Policies and Conditional Access Outcomes

The seven compliance policies define what “trusted enough to access corporate resources” means for managed mobile devices. They assess conditions such as minimum operating-system version, rooted or jailbroken status, encryption, password posture, platform integrity, and threat level. When combined with Microsoft Entra Conditional Access, the resulting compliance state becomes an access-control signal so that only devices meeting the organization’s requirements can reach protected resources.

For Android fully managed devices, the package moves from baseline trust at Level 1 — password, encryption, Play integrity, minimum Android 9.0, Intune app integrity — to enhanced password lifecycle at Level 2 and threat-protection / Android 11.0 minimum at Level 3. For Android work profile devices, Level 2 establishes BYOD-oriented trust with rooting block, encryption, Google Play services/security providers, Company Portal integrity, unknown-source blocking, and USB debugging block; Level 3 adds threat protection, Android 11.0 minimum, and stronger password lifecycle controls.

For iOS/iPadOS, Microsoft’s framework has no basic compliance offering because the available compliance settings do not justify a separate Level 1. The imported Level 2 policy blocks jailbroken devices, requires passcode and simple-passcode blocking, and sets a minimum iOS/iPadOS version; Level 3 raises the OS requirement and integrates secured threat posture / Microsoft Defender for Endpoint or mobile threat defense signals.

[iOS App Protection Policy Settings](#)

Device Configuration Policies and Security Outcomes

The eleven configuration policies establish the device-side hardening baseline that compliance policies can assess and Conditional Access can enforce. Across Android Enterprise and iOS/iPadOS, the framework applies progressively stronger password, data-transfer, account, sharing, storage, application, and operating-system restrictions as users and devices move from Level 1 to Level 3.

For Android work profile devices, the package applies work/personal separation: copy/paste restrictions, work profile screen-capture blocking, account-add blocking, work profile password requirements, and stricter high-security controls such as work-notification blocking, contact-search blocking, widget restrictions, trust-agent blocking, and reduced wipe thresholds. Microsoft states Android work profiles create a separate partition for work apps/data while personal apps and data remain unaffected by Intune policies, which is important for BYOD privacy expectations.

For iOS/iPadOS personal devices, Level 1 establishes password, lock-screen privacy, encrypted backup, AirDrop unmanaged target behavior, iCloud managed-app sync blocking, and certificate trust restrictions; Level 2 adds managed document transfer controls and diagnostic/enterprise book restrictions; Level 3 adds screen-capture blocking, Handoff blocking, password history/expi-

ration, and stricter wipe thresholds. For supervised devices, Microsoft's guidance includes more administrative controls, and the imported Level 3 supervised policy uses the strongest restrictions such as App Store block, AirDrop block, account/profile modification block, VPN creation block, iCloud backup/document sync block, Siri/iTunes/game restrictions, AirPrint block, and app removal block.

[iOS/iPadOS supervised device security configurations examples](#)

[iOS/iPadOS personal device security configuration examples](#)

Recommended Deployment Model

A practical phased deployment model is to assign Level 1 broadly, move most managed mobile users to Level 2, and reserve Level 3 for high-risk roles and high-sensitivity data scenarios. Microsoft recommends treating the levels as starting points and customizing them to the organization's threat environment, risk appetite, and usability impact; the same guidance appears across the APP, Android, and iOS/iPadOS framework documentation.

Recommended assignment pattern:

1. Foundation / broad baseline: Deploy Level 1 App Protection to all mobile users and Level 1 device configuration where available, especially for corporate-owned Android and iOS/iPadOS supervised/personal baselines.
2. Standard enterprise posture: Assign Level 2 APP, compliance, and configuration to most users who access Microsoft 365 work data from managed or BYOD mobile devices.
3. High-risk posture: Assign Level 3 policies to sensitive user groups and devices, including executives, privileged administrators, regulated-data users, and users likely to be targeted by sophisticated adversaries.
4. Conditional Access enforcement: Use device-based Conditional Access for enrolled/compliant devices and app-based Conditional Access for managed-app access; Microsoft states Conditional Access works with Intune compliance, configuration, and app protection policies.
5. Pilot and ring deployment: Validate policy impact in rings before broad production assignment; Microsoft explicitly recommends a ring methodology for APP and similar testing/ring approaches for Android and iOS/iPadOS security frameworks.

Framework Outcomes and Rapid Implementation

The framework delivers a Microsoft-aligned mobile security baseline across Android Enterprise and iOS/iPadOS. Its outcome is a coordinated set of controls for app-level data containment, device trust and threat-posture evaluation, platform hardening, and Conditional Access enforcement. It supports both BYOD and corporate-owned scenarios through managed-app protection, Android work-profile separation, and iOS/iPadOS personal and supervised device controls.

For administrators, the framework provides a ready-to-adapt policy catalog that can be mapped to security groups, assignment filters, enrollment types, and Conditional Access policies. Although the underlying Microsoft guidance is established, implementing it manually can be time-consuming because the complete posture spans multiple policy families, platforms, ownership models, and protection levels.

The accompanying PowerShell script removes much of this repetitive technical work by rapidly creating the policy baseline in Intune. Administrators can then validate the imported settings, define assignments and exclusions, pilot the controls through rollout rings, and begin enrolling devices against the approved configuration. The automation accelerates availability of the result; governance decisions, testing, and production approval remain the responsibility of the implementing organization.